

Lecture 11

Subspace of 3×3 $M \rightarrow$

- Symmetric matrices
- Upper triangular.

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ 0 & \cdot & \cdot \\ 0 & 0 & \cdot \end{bmatrix} \quad 6 \text{ dimensional.}$$

Basis for M (3×3 matrices)

$$\dim(M) = 9 \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

• Symmetric 3×3 basis (S)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

③

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3

$$\underline{\underline{\dim(S) = 6}}$$

• Upper triangular (U)

$$\underline{\underline{\dim(U) = 6}}$$

Natural basis for m contains U's basis

• $S \cap U \rightarrow$ Diagonal matrices (D)

$$\dim(D) = 3$$

① Sum of subspaces.

$S + U \rightarrow$ Combination of the subspaces.

in this particular case $S + U$ would be equal to M .

$S + U$ spans M

$$\underline{\underline{\dim(S+U) = 9}}$$

$$\# \dim(S) + \dim(U) = \dim(S \cap U) + \dim(S+U)$$

$$6 + 6 = 3 + 9$$

$$\textcircled{12} = \textcircled{12}$$

$$\# \quad \frac{d^2 y}{dx^2} + y = 0 \quad [\text{vector space}]$$

$$y = \cos x, \sin x$$

$$y = a \cos x + b \sin x$$

$$a, b \in \mathbb{C}$$

Solutions for y form a vector space.

Basis $\rightarrow$ $\left[\sin x \right], \left[\cos x \right]$ $\textcircled{\times}$
 2 dimensional basis $\textcircled{\times}$
 $\rightarrow \left[\sin x \right]$ 1d basis

$$\sin^2 x + \cos^2 x = 1$$

$\Rightarrow$ Actually $[\sin x], [\cos x]$ are dependent but not linearly so they are perfectly valid basis.

So, 2 dimensional basis.

Rank 1

$$\bullet A = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 8 & 10 \end{bmatrix} \quad \text{Basis (row)} \rightarrow \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$$

$$\dim(A^T) = 1$$

$$\dim(C(A)) = 1$$

$$\dim(N(A)) = 2$$

$$\dim(N(A^T)) = 1$$

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$$

$$\text{Basis (col)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\text{Basis } (N(A)) = a \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix} + b \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix} \quad \underline{\underline{\dim = 2}}$$

All rank 1 matrices are in the
form $A = uv^T$

eg $M = 5 \times 7$ matrices

Subset of rank 4 matrices is a subspace?

$$\Rightarrow \begin{aligned} A_5 &= c_1 \underline{A_1} + c_2 A_2 + \dots + c_n A_n \\ B_5 &= b_1 \underline{B_1} + b_2 B_2 + \dots + b_n B_n \end{aligned}$$

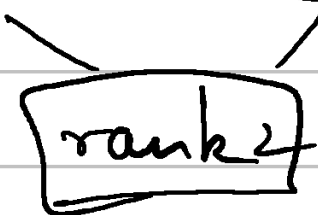
$A + B$ is not necessarily a
rank 4 as $c_i A_i + b_i B_i$
 is not necessarily going to make
 $A_6 + B_6$ dependent!

$$\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$$

eg subset of rank 1 matrices $\rightarrow$

Not a subspace as well.

eg $a_1 \begin{bmatrix} 1 & 0 \end{bmatrix}$ $a_2 \begin{bmatrix} 0 & 1 \end{bmatrix}$



eg $\underline{9m} \quad \underline{\mathbb{R}^4}$ $v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$

$S = \text{All } v \text{ in } \mathbb{R}^4 \text{ with } \sum_{i=1}^4 v_i = 0$

$\rightarrow v + u = \begin{bmatrix} v_1 + u_1 \\ v_2 + u_2 \\ \vdots \end{bmatrix} = \sum (v_i + u_i) = \sum v_i + \sum u_i = \underline{\underline{0}}$

$\lambda v = \checkmark$

Yes, it is a subspace!

dimension = 3

basis $\Rightarrow \begin{bmatrix} 1 \\ -1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$ intuition

$\boxed{A}v = 0$

$A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$

So, in essence, we want the null space of A .

$$\begin{bmatrix} \textcircled{1} & 1 & 1 & 1 \end{bmatrix}$$

pivot free

$$\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow \text{Another basis}$$

• Basis of $N(A)$

• Basis for $C(A) \Rightarrow \underline{\underline{[1]}} \quad \boxed{\mathbb{R}^1}$

• Basis for row space $\Rightarrow [1 \ 1 \ 1 \ 1]$
there is only one row

• Basis for $N(A^T)$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} m &= 4 \\ r &= 1 \\ m - r &= 0 \end{aligned}$$

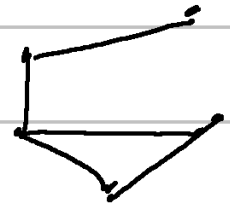
only {0}

inh

The only subspace with zero dimension is the $\{0\}$ null space and the basis for that is the empty set

Small World Graphs

Graph $\rightarrow$ nodes & edges.



$\Rightarrow$ How many steps for nodes as people & edges as friends for furthest apart.

The concept of degrees of separation